JMN Journal of Medical Sciences | Volume 1, Issue 1, Jul–Dec 2025 | pp. 3–10

Original Article

A Study on Urinary Incontinence and Its Risk Factors Among Post-menopausal Women in a Rural Area of West Bengal

Sabyasachi Roy[1*], Chirasree Sarkar[2]

[1]Associate Professor, Department of Community Medicine, JMN Medical College, Chakdaha
[2]Senior Resident, Department of Community Medicine, JMN Medical College, Chakdaha

Abstract

Background: Urinary incontinence (UI) is the involuntary loss of urine which is objectively demonstrable and is a social and hygienic problem. UI is a chronic debilitating disease which is often underreported, but has significant impact on daily activities, and is thus of public health importance. **Objectives:** To find out the proportion of rural post-menopausal women having UI and its associated risk factors. **Methods:** A cross-sectional study was conducted in a rural area of Barasat among 120 elderly women ($\geq$50 years) selected by cluster sampling (8 subjects each from 15 clusters) with a pre-designed questionnaire. Chi-square test and logistic regression were applied to find associations between variables. $P<0.05$ was considered statistically significant. Data were analyzed with SPSS version 16.0. **Results:** 43 (35.8%) of 120 women were found to have UI. The most prevalent type was stress UI (18.3%), followed by mixed UI (10.8%) and urge UI (5.8%). On multivariable analysis, illiteracy (AOR 2.41 [1.02–5.69]), NVDs >3 (AOR 3.37 [1.54–7.37]), history of gynaecological operation (AOR 3.84 [1.16–12.66]), chronic cough (AOR 2.69 [1.21–5.99]), and LUTS (AOR 2.63 [1.15–6.00]) remained significantly associated with UI. Only 25 (20.8%) sought medical help. **Conclusion:** Rural women are at high risk of developing UI, and most affected women did not seek treatment, which is a matter of concern. Awareness generation regarding UI may help improve health-seeking behaviour.

Key Words: Lower urinary tract symptoms, Post-menopausal women, Risk factors, Rural health, Urinary incontinence

Corresponding Author: Dr. Sabyasachi Roy, Department of Community Medicine, JMN Medical College, Chakdaha

Ethics Committee Clearance: Obtained from the Institutional Ethics Committee

Source(s) of Support: Not stated in the submitted manuscript.

Conflict of Interest: Not stated in the submitted manuscript.

How to cite this article: Roy S, Sarkar C. A study on urinary incontinence and its risk factors among post-menopausal women in a rural area of West Bengal. JMN J Med Sci. 2025 Jul–Dec;1(1):3–10.

I. Introduction

Urinary incontinence is a common distressing problem amongst adults living in the community [1]. Control of the micturition reflex is a complex physiological and anatomical process that may often fail, particularly in women. According to the International Continence Society (ICS), urinary incontinence (UI) is defined as: the complaint of any involuntary loss of urine that is a social or hygienic problem [2]. Worldwide, it has been observed that over 200 million people are living with urinary incontinence that may hamper their normal physical activity [3]. It is three times more common in women than men [4]. Urinary incontinence is a troublesome and probably underreported disorder, especially in women [5]. An estimated 19% of women under the age of 45 experience UI and 29% of women over the age of 80 also experience this condition [6] . In general, the overall prevalence of UI increases with age

[7]. Numerous epidemiologic studies show that the incidence of UI increases with age, with the range of prevalence estimates among community dwelling patients varying enormously from 2% to 58%[8]. Postmenopausal or elderly women are the most affected with a mean prevalence of 34% [9]. Women suffering from urinary incontinence constitute about 70-80% of female patients reporting to a physician due to urinary system diseases [10]. It is estimated that women of different ages have involuntary urine loss and specially 9-39% of women over 60 years report daily urinary leakage [11,12]. The prevalence of urinary incontinence is gradually increasing and may diminish quality of life and disrupt their daily routines [13]. Recently, as the average life expectancy of women has increased and an increasing number of women maintain various social roles, urinary incontinence (UI) has become an important medical and social issue [14]. Urinary continence is due to the fact that bladder pressure remains lower than urethral closure pressure, and urinary incontinence may result from bladder or urethral impairment; when closure pressure is lower than bladder pressure, urinary leakage occurs [15] . Among elderly women, many are in the postmenopausal phase; so the low estrogen production produces symptoms from the autonomic nervous system such as flushing and perspiration and psychiatric symptoms such as depression. Furthermore, the mucous membranes of the genitourinary tract among the elderly women undergo atrophy resulting in vaginitis, dyspareunia and cysto-urethritis [16]. Basically there

are three types of incontinence : stress urinary incontinence (SUI); urge urinary incontinence (UUI), and mixed urinary incontinence (MUI), which associates the first two [17] . Stress is most common occurring in pure or mixed forms, in 80% women with incontinence defined as the complaint of involuntary leakage on effort or exertion, or on sneezing or coughing. [11,14,16]

. Urinary incontinence hampers the person's freedom and ability to function in everyday life. It has therefore become a public health problem, with a high economic and human impact [20]. Although the interest is growing regarding the investigation, treatment, and management of incontinence, its prevalence in the general population has so far been truly based on some estimates made upon selected groups of people of different ages [21]. Despite its high prevalence, very little is known about risk factors associated with UI, and the etiology of urinary incontinence is poorly noted. Various factors which are associated with lifestyle are thought to aggravate lower urinary tract symptoms and results in urinary incontinence, for example, ageing, pregnancy, vaginal delivery, obesity physical forces (exercise, work), smoking, caffeine and fluid intake, constipation, posture [15,18,19] . Therefore, this study was carried out to estimate the prevalence of Urinary Incontinence among the postmenopausal (>60 years) women and to find out the risk factors among them associated with Urinary Incontinence. Most studies on urinary incontinence have been carried out in developed countries and affluent study populations. Scarce data exist on its prevalence in India, specially in rural women, or in those with lower socioeconomic status. The condition is usually under reported as most of the women hesitate to seek help or report symptoms to medical practitioners due to the embarrassing and culturally sensitive aspect of this condition. So if we can achieve

to get an updated picture of UI among rural women, that will be of great importance in helping formulating strategies of prevention and control for UI and that further helps to reduce the disease burden in India . In this aspect this study examined prevalence of UI among rural women, and the risk factors associated with it among the study participants.

II. Materials and Methods

A community-based cross-sectional study was conducted from May to July 2022 in the rural area of Barasat, West Bengal comprising 64 villages. All the postmenopausal women(>=60 years) [22] in the study setting were included, but those who were critically ill patients or who were not willing to participate in the study or those with known psychiatric illness they were excluded from the study. Sample size will be calculated

using (Z2.P.Q/L2) as per cochran's formula. Where, P = Prevalence of cardiovascular risk, Z=

1.96 (95% confidence interval), L= absolute error of precision. Considering the following assumptions prevalence of UI among postmenopausal (>60 years) women was found to be 27.7% [23] among women in rural India from a similar study. Taking L = 10%, sample size(n) came= 75.71. Considering a design effect of 1.5 for cluster sampling, it came =(75.51 $\times$ 1.5)= 113 and taking 10% non-response rate, final sample size came 120. A two stage cluster sampling was done to select study participants. In study area there were 64 villages, from these villages at first stage 12 villages or cluster were selected randomly. With the help of local maps and local people after going to the centre of the first village, at first one direction was chosen randomly by rotating bottle head. Going to that direction one house was chosen. That was the first house to elderly women participant as per inclusion criteria. Next adjacent houses were visited until 10 participants were selected (At the second stage 10 elderly women were selected from each cluster (120/12=10). When there was end of road, next lane was taken to complete sample size. Then the same procedure was repeated in rest of all 11 clusters. [Fig 1]. Face to face interview of the participants were conducted with the pre-designed, pre-tested questionnaire and anthropometric and clinical examination was done. Urinary Incontinence, dependent variable was assessed by QUID Questionnaire [24]) was the dependent variable. Other independent variables were sociodemographic variable- Age in completed years, Level of Education (in completed years of Education), Religion, Caste, Family type, Socio economic class of the family (Modified B.G. Prasad Scale, 2020), Occupational Status ; reproductive variables -Menstrual cycle regularity, no of child birth, h/o abortion/PCOD, age of first child birth, age of marriage, marital status, complication during pregnancy, ever OCP usage; behavioural variables-Tobacco/Alcohol use, Physical Activity Status assessed by IPAQ- SF[25] (International Physical Activity Questionnaire- short form); Mental stress(assessed by PSS-4); others- H/O chronic illness, family h/o any chronic illness(Hypertension, Diabetes, Dyslipidemia), dietary Behaviour, BMI(body mass index). During data collection the past medical records of the participants were also assessed.

Ethical clearance was obtained. The study's purpose was described, and all participants gave their informed consent. Confidentiality and privacy of the participants were genuinely maintained throughout the whole study. Data were analyzed by Microsoft Excel 2019 and statistical package for social sciences (Version 16). Univariate logistic regression and multivariable logistic regression were done. P<0.05 was considered as statistically significant within 95% confidence interval.

III. Results

Majority of the study participants (44.2%) belonged to age group 70 years or above. The mean (standard deviation) of the age of the study participants was 60.88 ($\pm$7.65). Majority (95%) of the study participants were Hindu. About 59.2% participants were with no formal education. Most of the study participants (64.2%) belonged to others caste. Among the post menopausal age group women majority (61.6%) belonged to socio economic class 4.

About 78.3% were homemaker by occupation. Nearly 64.2% of post-menopausal age group women stay in the joint family. Nearly 32.5% of them were diabetic while 36.7% of them were hypertensive. [Table 1].

Nearly 32.5% of them were diabetic while 36.7% of them were hypertensive. Among the study participants 36.7% had chronic cough and 35% had h/o constipation. Nearly 22.5% of them had a history of gynecological operation. Among them 53.3% had more t han equals t o normal vaginal delivery (NVD), while 48.3% had a history of abortion . Around 25.8% had h/o fecal incontinence. About 37.5% of them were suffering from any one of the LUTSs. Among them 40% elderly women had low physical activity and 35% with high mental stress. According to BMI 25.8% were overweight and 2.5% were obese. [Table 2].

Forty-two (35%) out of 120 women were found having UI. The most prevalent type of UI was stress UI (18.3%);followed by mixed UI (10.8%) and urge UI (5.8%). It has been seen that 44.2% of post-menopausal age group aged 70years and above; 32.5% of aged 65- 69 years post-menopausal age group women and 23.3% post-menopausal age group women aged 60-64 years had urinary incontinence. Only 15 (30.6%) sought treatment for UI.

In bivariate logistic regression analysis, study participants, who were illiterate, having a history of gynecological operation, NVDs (>3), diabetic, hypertension, chronic cough, constipation, LUTS, had shown significantly greater odds of having UI. In multivariable model hypertension (adjusted odds ratio [AOR] - 2.15 [1.13-4.75]), chronic cough (AOR

- 4.50 [1.24-16.30]), and constipation (AOR - 8.58 [2.06-35.65]) remained significant adjusted with other significant variable in bivariate analysis [Table 3].

Table 1: *Distribution of study participants according to background characteristics (n=120)*

Parameters	Category	n (%)
Age (years)	60–64	28 (23.3)
	65–69	39 (32.5)
	≥70	53 (44.2)
Religion	Hindu	114 (95.0)
	Muslim	6 (5.0)
Educational level	Illiterate	71 (59.2)
	Below primary	14 (11.7)
	Primary	25 (20.8)
	Middle	6 (5.0)
	Secondary	4 (3.3)
Caste	General/Others	77 (64.2)
	SC	43 (35.8)
	OBC	5 (4.2)
Socioeconomic class	Class 1	2 (1.6)
	Class 2	5 (4.1)
	Class 3	17 (14.1)
	Class 4	74 (61.6)
	Class 5	22 (18.3)
Occupation	Homemaker	94 (78.3)
	Cultivator/Biri binder	26 (21.7)
Family type	Nuclear	43 (35.8)
	Joint	77 (64.2)

Table 2: *Distribution of study participants according to other characteristics (n=120)*

Parameters	Category	n (%)
Diabetes	Absent	81 (67.5)
	Present	39 (32.5)
Hypertension	Absent	76 (63.3)
	Present	44 (36.7)
Chronic cough	No	76 (63.3)
	Yes	44 (36.7)
Constipation	No	78 (65.0)
	Yes	42 (35.0)
NVD	≤3	64 (53.3)
	>3	56 (46.7)
LUTS	No	75 (62.5)
	Yes	45 (37.5)
H/O gynae. operation	No	93 (77.5)
	Yes	27 (22.5)
Faecal incontinence	No	89 (74.2)
	Yes	31 (25.8)
H/O abortion	No	58 (48.3)
	Yes	62 (51.7)
BMI	Underweight (<18.5)	2 (1.7)
	Normal (18.5–24.9)	84 (70.0)
	Overweight (25–29.9)	31 (25.8)
	Obese (≥30)	3 (2.5)
Physical activity	Low	48 (40.0)
	High	72 (60.0)
Mental stress	Low	78 (65.0)
	High	42 (35.0)

Table 3: *Factors associated with presence of urinary incontinence among study participants: logistic regression analysis (n=120)*

Variable	Category	Total n (%)	UOR (95% CI)	AOR (95% CI)
Diabetes	Absent	81 (67.5)	1 (Ref)	1 (Ref)
	Present	39 (32.5)	12.18 (4.93–30.09)	2.28 (0.56–9.26)
Hypertension	Absent	76 (63.3)	1 (Ref)	1 (Ref)
	Present	44 (36.7)	14.09 (5.67–34.99)	2.15 (1.13–4.75)
Chronic cough	No	76 (63.3)	1 (Ref)	1 (Ref)
	Yes	44 (36.7)	14.09 (5.67–34.99)	4.50 (1.24–16.30)
Constipation	No	78 (65.0)	1 (Ref)	1 (Ref)
	Yes	42 (35.0)	21.76 (8.23–57.31)	8.58 (2.06–35.65)
LUTS	No	75 (62.5)	1 (Ref)	1 (Ref)
	Yes	45 (37.5)	6.00 (2.63–13.65)	0.73 (0.17–3.01)
NVD	≤3	64 (53.3)	1 (Ref)	1 (Ref)
	>3	56 (46.7)	8.76 (3.62–21.20)	2.35 (0.62–8.83)
H/O gynae. op.	No	93 (77.5)	1 (Ref)	1 (Ref)
	Yes	27 (22.5)	4.62 (1.86–11.43)	0.52 (0.09–2.73)

UOR: Unadjusted odds ratio; AOR: Adjusted odds ratio; CI: Confidence interval

IV. Discussion

About 35% of the study population (postmenopausal age group) was found to have UI. The findings were consistent with findings of Prabhu and Shanbhag[26] (25.5%), Ansar et al.[27] (23.9%), and Seshan and Muliira[28] (33.8%), Abha et al.[29] (34.0%), Kılıç[30] (37.2%), and Sensoy et al.[31] (44.6%) have reported like this. Singh et al.[32] (21.8%), Ge et al.[33] (22.1%), Sumardi et al.[34] (13.0%), Abiola et al.[35] (12.6%), and Bodhare et al.[36] (10.0%) have reported less. This variance in the prevalence may be due to different study population, settings, and definition of UI used.

The study revealed that illiterate women are at more risk. Hence educational level plays a vital role in this regard. Sensoy et al.,[31] Seshan and Muliira,[28] and Ge et al.[33] had similar findings. A history of prolonged labor: it is an established cause of urinary tract injury-thus increases the risk of UI. The study had identified a history of prolonged labor as a risk factor for UI similarto Bodhare et al.[36] and Kılıç[30] with increase in number of childbirth chance of trauma to the urinary tract also increases. The study established NVD as a risk factorfor UI similar as Singh et al.,[32] Ge et al.,[33] Seshan and Muliira,[28] and Sensoy et al.[31]

A history of gynecological operation imposes risk of iatrogenic trauma to the urinary tract, thus increasing risk ofUI. The study revealed that also which is similar to findingsof Prabhu and Shanbhag,[26] Sensoy et al.,[31] Ge et al.,[33]and Singh et al.[32] Diabetes causes polyuria which laid additional burden on sphincters of urinary tract resulting in UI. The current study also establishes the fact same as Prabhu and Shanbhag,[26] Kılıç,[30] Ge et al.,[33] and Singh et al.[32] Constipation and chronic cough create additional stress on both the anal and urethral sphincter resultinginto UI. The studies by Prabhu and Shanbhag,[26] Bodhareet al.,[36] and Sensoy et al.[31] identified constipation and chronic cough both as risk factors of UI while Ge et al.[33]found constipation and Sumardi et al.[34] identified chroniccough as a risk factor. The study establishes LUTS as oneof the most important risk factors for UI similar to Sumardiet al.,[34] Sensoy et al.,[31] Ge et al.,[33] and Kılıç.[30]

The most prevalent type of UI was stress UI (18.3%), followed by mixed UI (10.8%) and urge UI (5.8%).This finding further corroborated earlier reports from epidemiological studies of UI in India in which stress incontinence was the most common type of UI among women with UI.[26,29,32] . Only 30.6% sought treatment for UI which is better compared to Prabhu and Shanbhag[26] (14.4%), Sarici et al.[37] (10.7%), Jokhio et al.[38] (15.7%) and worsecompared to Lasserre et al.[39] (39.7%).

V. Conclusion

Prevalence of UI is high in rural postmenopausal age group women. Most prevalent one is stress UI. Majority of them did not sought treatment for UI which is matter of concern. Generating awareness regarding UI may help to improve health-seeking behavior. In the current study, UI is predicted by certain modifiable risk factors, proper management of which is warranted to reduce the burden of UI and thus improving the health status of elderly women. Several treatment choices for UI are now available with greater effectiveness. For example, pelvic floor muscle exercise or perineal exercise is established to be effective first-line intervention to prevent the development of UI and further complication. Thus, encouraging pelvic floor muscle exercise or perineal exercise among the at risk women and those who are affected is useful in both the prevention and management of UI. An attempt

should be made by doctor or health personnel to screen at risk women for UI at every given opportunity by asking leading questions so that the affected can be timely referred for relevant management.

References

[1] Jolleys JV. Reported prevalence of urinary incontinence in women in a general practice. Br MedJ (Clin Res Ed). 1988;296:1300.

[2] Hsieh CH, Su TH, Chang ST, Lin SH, Lee MC, Lee MY. Prevalence of and attitude toward urinary incontinence in postmenopausal women. International J Gynecol Obstet. 2008;100(2):171-4.

[3] Jackson SL, Scholes D, Boyko EJ, Abraham L, Fihn SD. Urinary incontinence and diabetes in postmenopausal women. Diab Care. 2005;28(7):1730-8.

[4] Jameel S, Mahmud SN. Frequency of different risk factors associated with Recurrent Urinary Tract Infection among Postmenopausal Women. J AyubMed Coll Abbottabad. 2016;28(2):353-6.

[5] Wyman J. The psychiatric and emotional impact of female pelvic floor dysfunction. Curr Opin Obstet Gynecol. 1994;6:336-9.

[6] Hollywood B, O'Dowd T. Female urinary incontinence: another chronic illness. Br J Gen Pract.1998;48:1727-8.

[7] Grimby A, Milson I, Molander U et al. The influenceof urinary incontinence on the quality of life of elderly women. Age Ageing. 1993;22:82-9.

[8] Lam G, Foldspang A, Elving LB, Mommsem S. Social context, social abstention, and problem recognition correlated with adult female urinary incontinence. Dan Med Bull. 1992;39:565-70.

[9] Bogner HR, Gallo JJ, Sammel MD, et al. Urinary incontinence and psychological distress in community-dwelling older adults. J Am Geriatr Soc. 2002;50:489-95.

[10] Bai SW, Jeon MJ, Kim JY, Chung KA, Park KH. Int Urogynecol J Pelvic Floor Dysfunct. 2002; 13(4):256-60.

[11] Jarvis GJ. The place of urodynamic investigation in female urinary incontinence. London: Royal College of Obstetricians and Gynaecologists,1990.

[12] Stewart WF, Van Rooyen JB, Cundiff GW, Abrams P, Herzog AR, Corey R et al. Prevalence and burden of overactive bladder in the United States. World J Urol. 2003;20:327-36.

[13] Melville JL, Walker E, Katon W et al. Prevalence of comorbid psychiatric illness and its impact onsymptom perception, quality of life, and functional status in women with urinary incontinence. Am J Obstet Gynecol. 2002.187:80-7.

[14] Shaw C, Tansey R, Jackson C et al. Barriers to help seeking in people with urinary symptoms. Fam Pract.2001;18:48-52.

[15] Pereira VS, de Melo MV, Correia GN, Dri usso P. Long-term effects of pelvic floor muscle training with vaginal cone in postmenopausal women with urinary incontinence: a randomized controlled trial. Neurourol Urodynam. 2013;32:48-52.

[16] Oyen HV and Oyen PV. Urinary Incontinence inBelgium; Prevalence, Correlates and Psychosocial Consequences. Acta Clinica Belgica. 2002;57(4).

[17] Kelleher CJ, Cardozo LD, Khullar V. A newquestionnaire to assess the quality of life of urinary incontinent women. Br J Obstet Gynaecol, 1997.

[18] Kirss F, Lang K, Toompere K, Veerus P. Prevalence and risk factors of urinary incontinence among Estonian postmenopausal women. Spring Plus. 2013;2:524.

[19] Milsom I, Abrams P, Cardozo L, Roberts RG, Thuroff J, Wein AJ et al. How widespread are the symptoms of an overactive bladder and how are they managed? A population-based prevalence study. BJUInt. 2001;87(9):760-6.

[20] Brown JS, Vittinghoff E, Wyman JF, Stone KL, Nevitt MC, Ensrud KE et al. Urinary incontinence: does it increase risk for falls and fractures? Study of osteoporotic fractures research group. J Am Geriatr Soc. 2000;48(7):721-5.

[21] Tsai YC, Liu CH. Urinary incontinence among Taiwanese women: an outpatient study of prevalence, comorbidity, risk factors, and quality of life. Int Urol Nephrol. 2009;41:795-803.

[22] Ministry of Statistics and Programme Implementation. Elderly in India 2016.India: Government of India; February 2016.104p.

[23] Biswas B, Bhattacharyya A, Dasgupta A, Karmakar A, Mallick N, Sembiah S. Urinary incontinence, its risk factors, and quality of life: a study among women aged 50 years and above in a rural health facility of West Bengal. Journal of mid-life health. 2017 Jul;8(3):130.

[24] Ajith, Aathira Kizhakkeveetil et al. "Prevalence and Factors of Urinary Incontinence among Postmenopausal Women Attending the Obstetrics and Gynecology Outpatient Service in a Tertiary Health Care Center in Kochi, Kerala." Indian journal of community medicine : official publication of Indian Association of Preventive & Social Medicine vol. 44,Suppl 1 (2019): S30-S33. doi:10.4103/ijcm.IJCM_29_19

[25] Ahmad, Mohamad Hasnan et al. "Comparison between self-reported physical activity (IPAQ-SF) and pedometer among overweight and obese women in the MyBFF@home study." BMC women's health vol. 18,Suppl 1 100. 19 Jul. 2018, doi:10.1186/s12905- 018-0599-8

[26] Prabhu SA, Shanbhag SS. Prevalence and risk factors of urinary incontinence in women residing in a tribal area in Maharashtra, India. J Res Health Sci 2013;13:125-30.

[27] Ansar H, Adil F, Munir AA. Unreported urinary and anal incontinence in women. J Liaquat Univ Med Health Sci 2005;4:54-6.

[28] Seshan V, Muliira JK. Self-reported urinary incontinence andfactors associated with symptom severity in community dwelling adult women: Implications for women's health promotion. BMC Womens Health 2013;13:16.

[29] Abha S, Priti A, Nanakram S. Incidence and epidemiologyof urinary incontinence in women. J Obstet Gynecol India 2007;57:155-7.

[30] Kılıç M. Incidence and risk factors of urinary incontinence in women visiting family health centers. Springerplus 2016;5:1331

[31] Sensoy N, Dogan N, Ozek B, Karaaslan L. Urinary incontinence in women: Prevalence rates, risk factors and impact on quality of life. Pak J Med Sci 2013;29:818-22.

[32] Singh U, Agarwal P, Verma ML, Dalela D, Singh N, Shankhwar P, et al. Prevalence and risk factors of urinary incontinence in Indian women: A hospital-based survey. Indian J Urol 2013;29:31-6.

[33] Ge J, Yang P, Zhang Y, Li X, Wang Q, Lu Y, et al. Prevalence and risk factors of urinary incontinence in Chinese women: A population-based study. Asia Pac J Public Health 2015;27:NP1118-31.

[34] Sumardi R, Mochtar CA, Junizaf, Santoso BI, Setiati S, Nuhonni SA, et al. Prevalence of urinary incontinence, risk factors and its impact: Multivariate analysis from indonesian nationwide survey. Acta Med Indones 2014;46:175-82.

[35] Abiola OO, Idowu A, Ogunlaja OA, Williams-Abiola OT, Ayeni SC. Prevalence, quality of life assessment of urinary incontinence using a validated tool (ICIQ-UI SF) andbothersomeness of symptoms among rural community: Dwelling women in Southwest. Nigeria 2016;3:989-97.

[36] Bodhare TN, Valsangkar S, Bele SD. An epidemiological studyof urinary incontinence and its impact on quality of life among women aged 35 years and above in a rural area. Indian J Urol 2010;26:353-8.

[37] Sarici H, Ozgur BC, Telli O, Doluoglu OG, Eroglu M, Bozkurt S, et al. The prevalence of overactive bladder syndrome and urinary incontinence in a Turkish women population; associated risk factors and effect on quality of life. Urologia 2016;83:93-8.

[38] Jokhio AH, Rizvi RM, Rizvi J, Macarthur C. Urinary incontinence in women in rural Pakistan: Prevalence, severity, associated factors and impact on life. BJOG 2013;120:180-6.

[39] Lasserre A, Pelat C, Guéroult V, Hanslik T, Chartier-Kastler E, Blanchon T, et al. Urinary incontinence in French women: Prevalence, risk factors, and impact on quality of life. Eur Urol 2009;56:177-83.